Compactness, incompactness, and determinacy

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How **strong** is the assumption that the universe of sets is **totally compact**?

Two themes from set theory

1. Compactness versus incompactness
2. Measuring the strength of extensions of ZFC

Infinite cardinal numbers

$$\aleph_0 < \aleph_1 < \aleph_2 < \cdots < \aleph_\omega = \sup\{\aleph_n : n < \omega\} < \aleph_{\omega+1} < \cdots < \aleph_\alpha < \ldots$$

The **cofinality** of a cardinal κ is the number of smaller cardinals needed to build κ

A cardinal κ is

- **singular** if $\mathrm{cof}(\kappa) < \kappa$
- **regular** if $\mathrm{cof}(\kappa) = \kappa$
- a **strong limit** if $2^\gamma < \kappa$ for all $\gamma < \kappa$.

Compactness and incompactness

Definition

Let κ be an uncountable cardinal. Then $\square_\kappa$ holds if there is a sequence $\langle C_\alpha : \alpha < \kappa^+ \rangle$ such that for all $\alpha < \kappa^+$

1. C_α is a closed, unbounded subset of α,
2. the order type of C_α is at most κ, and
3. if β is a limit point of C_α, then $C_\alpha \cap \beta = C_\beta$.

Compactness and incompactness

Used to build pathological objects

Theorem (Jensen)

$\square_\kappa$ implies there is a κ-Suslin tree and a nonreflecting stationary subset of κ

Contradicted by "reflection principles"

Theorem (Solovay)

If κ is a supercompact cardinal, then $\neg\square_\lambda$ for all $\lambda \geq \kappa$.

Theorem (Todorcevic)

The Strong Reflection Principle implies $\neg\square_\kappa$ for all uncountable κ.

Compactness and incompactness

Compactness/incompactness phenomena are studied using **forcing** and **large cardinal axioms**

Theorem (Cummings-Foreman-Magidor)

Suppose there is a supercompact cardinal. Then there is a forcing extension such that

1. $\square_{\aleph_\omega}$ *fails, and*
2. *in a further forcing extension,* $\square_{\aleph_\omega}$ *holds.*

Theorem (Cummings-Foreman-Magidor)

If there are infinitely many supercompact cardinals, then there is a forcing extension in which a weak version of $\square_{\aleph_\omega}$ *holds and stationary subsets of* $\aleph_{\omega+1}$ *reflect.*

Commitment to the higher infinite

The universe is totally **incompact** unless we assume (the consistency of) sufficiently strong large cardinals.

Theorem (Jensen)

In Gödel's Constructible Universe, square holds at all cardinals.

Commitment to the higher infinite

Inner model theory builds minimal models ("fine structural inner models") of large cardinal axioms

Theorem (Schimmerling-Zeman)

Suppose M is a fine structural inner model. Then in M, square fails at exactly the subcompact cardinals.

- Conditional: Inner models with subcompact cardinals have yet to be constructed

There are inner models for axioms up to "there is a Woodin cardinal which is a limit of Woodin cardinals"

- weaker than subcompact $<$ superstrong $<$ supercompact

Commitment to the higher infinite

Inner model theory is used to measure the consistency strength of combinatorial principles, especially failures of square

Example (Jensen-Schimmerling-Schindler-Steel)

$\mathrm{ZFC} + \aleph_2^\omega = \aleph_2 + \neg\square_{\aleph_2} + \neg\square_{\aleph_3}$ implies there is a model of ZFC with arbitrarily large Woodin cardinals and strong cardinals.

Corollary

The Strong Reflection Principle implies the same.

Road map

Compactness through $\aleph_\omega$

The Axiom of Determinacy

Nairian models

A totally compact universe

Compactness in the $\aleph_n$'s and $\aleph_\omega$

Theorem (Zeman)

1. *If there is a subcompact cardinal, then there is a forcing extension in which forms of stationary reflection hold at $\aleph_2$ and $\aleph_3$.*
2. *If there is a measurable subcompact cardinal, then there is a forcing extension in which $\square_{\aleph_\omega}$ fails.*

- Reduced hypothesis from supercompactness
- Techniques do not seem to generalize to e.g. singulars of uncountable cofinality

The prevailing belief

Conjecture (Steel)

Failure of square at a singular strong limit cardinal is at least as strong as a superstrong cardinal.

Prevailing belief

Total compactness "should" require the consistency of a supercompact cardinal.

Why is the problem hard?

If true: There is no general technique (yet) for showing that a theory requires the consistency of a superstrong cardinal.

- The conjectures are target problems for developing new methods

If false, and failures of square at the $\aleph_n$'s are sufficiently weak, then certain constructions which should produce an inner model with a subcompact cardinal do not provably succeed

- Consistent failure of a family of "iterability conjectures"

Our work shows that the previously known consistency upper bounds were far from optimal

Key idea: Changing the forcing ground model

Forcing over ground models of ZF + the **Axiom of Determinacy** (AD) can achieve effects only obtained in the standard way (using a ground model of ZFC plus large cardinals) from much stronger assumptions

Example

- (Woodin) The nonstationary ideal on ω_1 being ω_1-dense is consistent relative to the existence of infinitely many Woodin cardinals
- (Leitz) The same can be forced assuming a supercompact limit of supercompact cardinals

Key idea: Changing the forcing ground model

Guiding principle

Any statement about the $\aleph_n$'s which can be forced over a model of ZFC with large cardinals can be forced over a model of ZF + AD

- With a more optimal hypothesis

The Axiom of Determinacy

Let $\mathbb{R} = \omega^\omega$.

Let $A \subseteq \mathbb{R}$. Then G_A is the game in which players I and II alternate playing integers, producing $x \in \mathbb{R}$. Player I wins G_A if and only if $x \in A$.

The Axiom of Determinacy

For all $A \subseteq \mathbb{R}$, one of the players has a winning strategy in G_A.

- Implies all sets of reals are Lebesgue measurable
- Assumed to hold in inner models, not in the universe of sets

The Axiom of Real Determinacy ($\mathrm{AD}_\mathbb{R}$)

For all $A \subseteq \mathbb{R}^\omega$, G_A is determined.

"Standard" models of AD

Definition

Θ denotes the least ordinal α such that there is no surjection $\pi : \mathbb{R} \twoheadrightarrow \alpha$

- $|\mathbb{R}|^+$ in ZFC; very large assuming ZF+AD
- Strong models of AD are obtained by stipulating largeness properties of Θ
  - E.g., Θ is a regular cardinal

A "standard" model of AD is of the form $L(\wp(\mathbb{R}))$

- Incompact above its Θ

Forcing over standard models of AD

Martin's Maximum is a generalization of the Baire Category Theorem, and $\mathrm{MM}(c)$ is its restriction to partial orders of size at most $|\mathbb{R}|$

- $\mathrm{MM}(c)$ implies $\neg\square_{\aleph_1}$

Theorem (Woodin, 1990s)

There is a forcing extension of a standard model of $\mathrm{AD}_{\mathbb{R}}$ + *"Θ is regular" satisfying* $\mathrm{ZFC} + \mathrm{MM}(c)$.

Theorem (Caicedo-Larson-Sargsyan-Schindler-Steel-Zeman)

There is a forcing extension of a standard model of $\mathrm{AD}_{\mathbb{R}}$ + *"Θ is regular" satisfying* $\mathrm{ZFC} + \neg\square_{\aleph_1} + \neg\square_{\aleph_2}$.

Forcing over standard models of AD

The obstacle

Standard models of AD satisfy $\square_\Theta$

- Using the technique behind the Woodin and CLSSSZ theorems, Θ becomes $\aleph_3$ in the forcing extension

Need models of AD with compact structure at and above Θ

- But generalizing AD to higher types (sets of sets of reals, sets of... sets of reals) has been a longstanding, vague open problem

Failure of square at consecutive $\aleph_n$'s

Theorem (B.-Larson-Sargsyan)

Fix $n < \omega$. Suppose N is a model of $\mathrm{AD}_{\mathbb{R}}$ + "Θ is regular" in which the first $(n-3)$-many successors of Θ

1. *are regular cardinals in N and*
2. *of cofinality $\geq \aleph_2$ inside a larger model of AD.*[1]

Then there is a forcing extension of N in which square fails at $\aleph_1, \aleph_2, \ldots, \aleph_n$.

- **Key point**: Θ^{+i} is compact, and Θ^{+i} becomes $\aleph_{3+i}$ in the forcing extension

[1] Further requisite properties are suppressed :)

The transfinite generalization of AD

Definition

Assume Determinacy. Let λ be an ordinal. Then **the Nairian model at** λ is the structure

$$L_\lambda(\mathcal{H}, \bigcup_{\alpha<\lambda} \alpha^\omega),$$

where $\mathcal{H}$ is a predicate for a definable class of pairs of ordinals.

- Under the right Determinacy assumptions, Nairian models have rich structure above their Θ's

The transfinite generalization of AD

Theorem (B.-Larson-Sargsyan)

Nairian models satisfying $\mathrm{ZF} + \mathrm{AD}_{\mathbb{R}} +$ *"Θ is regular" exist assuming* **less than** *a Woodin limit of Woodin cardinals.*

Corollary

The failure of square at consecutive $\aleph_n$'s is **weaker** than a Woodin limit of Woodin cardinals.

Corollary

The "iterability conjectures" consistently fail.

The cost of total compactness

To push the forcing through the limit case, need a grip on the cardinal structure at and above $\Theta^{+\omega}$ in Nairian models

Theorem (B.-Larson-Sargsyan)

There is a forcing extension of a Nairian model in which square fails at all uncountable cardinals.

Corollary

The universe being totally compact is weaker than a Woodin limit of Woodin cardinals.

Structure of the argument

1. Define a theory T extending ZF with enough "reflection" to prove fragments of the Axiom of Choice after mild forcings
2. Show that the full Axiom of Choice can be "gently" forced over any model of T by wellordering powersets of cardinals
3. Show that Nairian models satisfy T
4. Show that the $\mathbb{P}_{\max}$-extension of a large enough Nairian model is closed under ω_1-sequences in the $\mathbb{P}_{\max}$ extension of an ambient determinacy model
   - To apply the argument that square fails throughout the model

Consequences

- The primary means for computing consistency lower bounds of (purportedly) strong principles has been failures of square principles
  - Either the principles are not strong, or they are out of reach for now
- It is consistent for certain inner model theoretic constructions to fail to produce an inner model with a subcompact cardinal

Towards more compactness

Theorem (B.-Larson-Sargsyan)

The Strong Reflection Principle holds in the $\mathbb{P}_{\max}$ extension of a Nairian model. Hence "weak square" fails for all singulars of countable cofinality.

- Previous best upper bound: a supercompact cardinal

In fact, essentially all named consequences of Martin's Maximum seem to hold in the extension.

Open problems

Question

Does Martin's Maximum hold in the extension of the Nairian model?

Question

What infinitary combinatorial statements require superstrong cardinals?

- Test candidate: The tree property at κ (every tree of height κ with levels of size $< \kappa$ has a cofinal branch)

Example (B.-Sargsyan-Trang)

There is a forcing extension of a Nairian model satisfying ZF + "every uncountable cardinal is singular"

- (Gitik) Previous bound: arbitrarily large strongly compact cardinals

Nairian models & the foundations of set theory

How strong can Nairian models be? How general is our technique/guiding principle?

Conjecture (Woodin)

Forcing extensions of Nairian models cannot contain Woodin cardinals

Answering this will tell us something about how Nairian models fit into the foundations of set theory

Thank you